

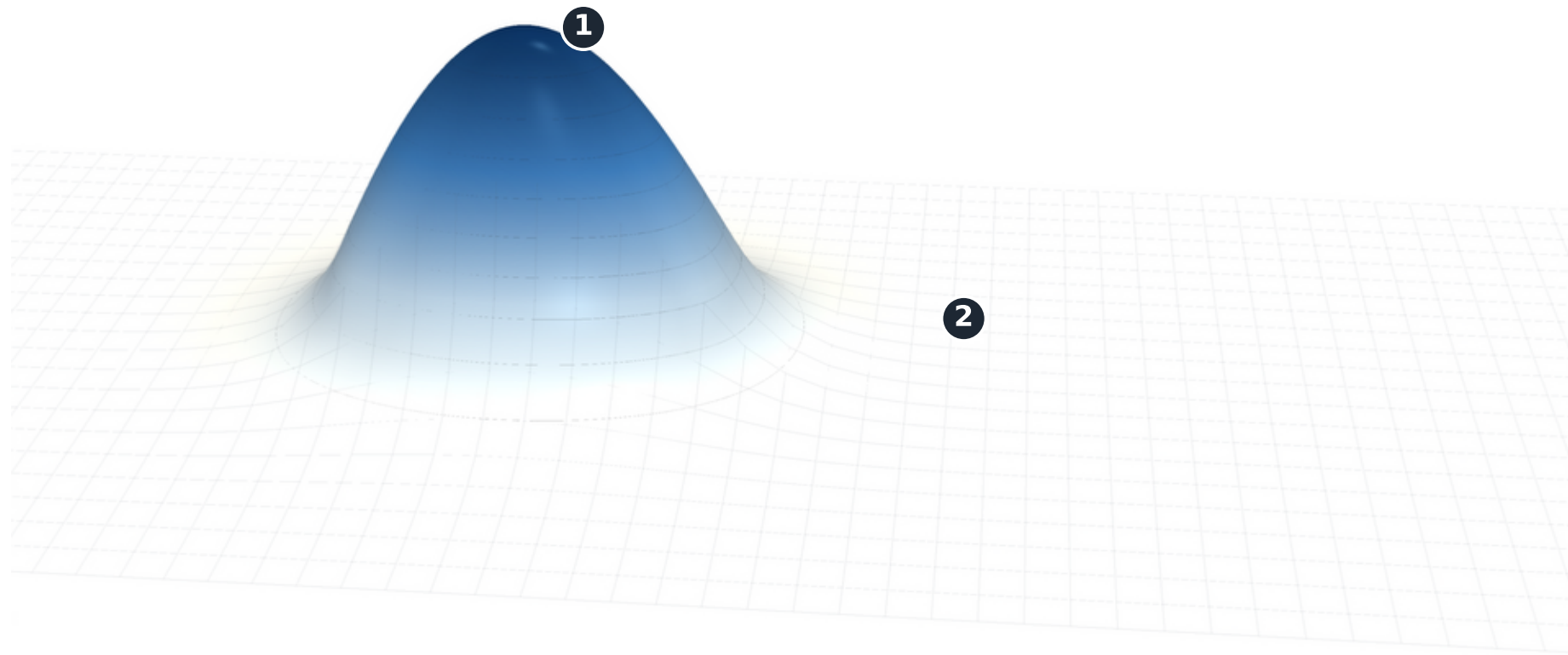
What does a single waveguide carry?

1. one guide

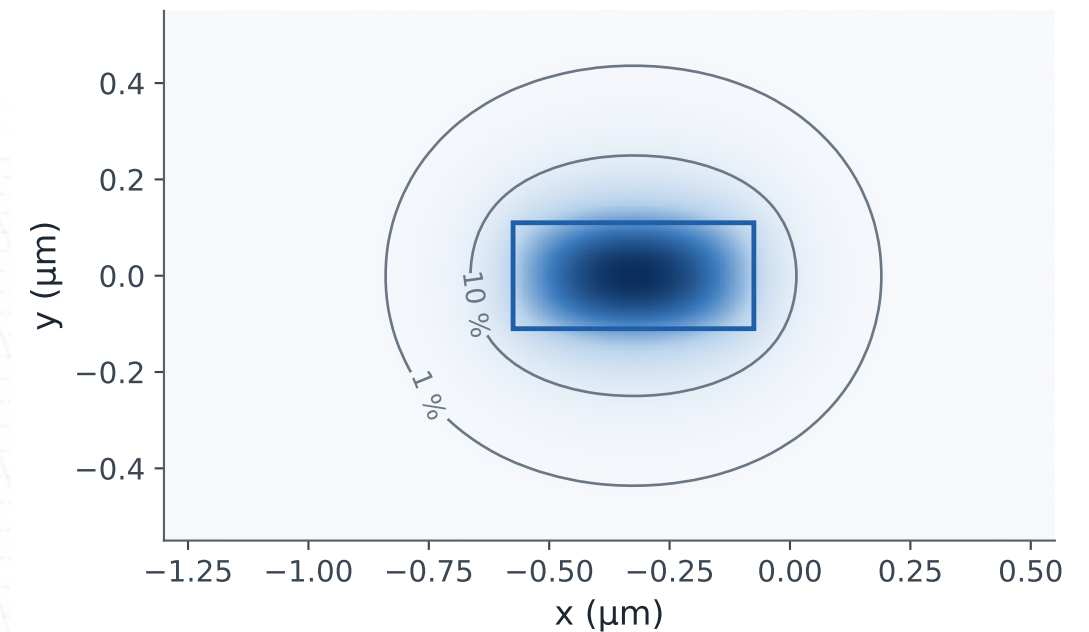
2. add B

3. A's tail
reaches B4. u_A no longer
exact5. two-mode
blend6. project
onto B7. the number
 κ_c 8. power
oscillates9. supermodes
beat10. 3-D Maxwell
check11. gap
sensitivity

mode shape $u_A(x, y)$ of guide A, drawn as height



the same mode, end-on



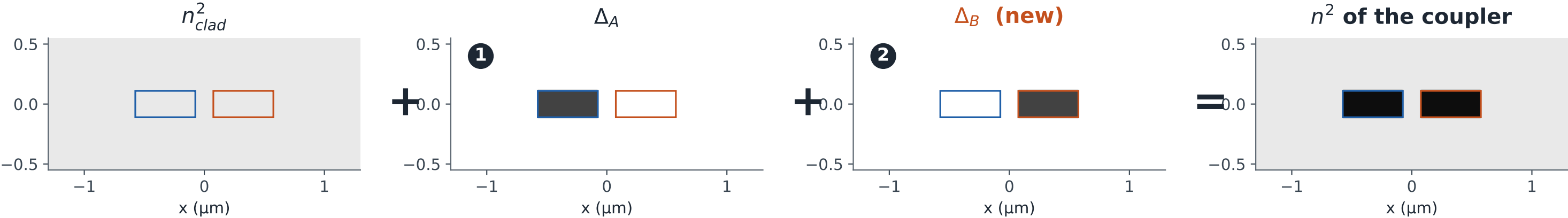
① The mode shape: how strong the field is across the guide. Solved from the scalar wave equation; $n_{\text{eff}} = 2.812$.

② The evanescent tail: the field does not stop at the silicon sidewall (contours: 10 % and 1 % of peak).

so → one guide carries one fixed shape that travels as $e^{-j\beta z}$, $\beta = 13.49$ rad/ μm . Nothing is coupled yet.

What changes physically when guide B is placed next to A?

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8. power oscillates
9. supermodes beat
10. 3-D Maxwell check
11. gap sensitivity

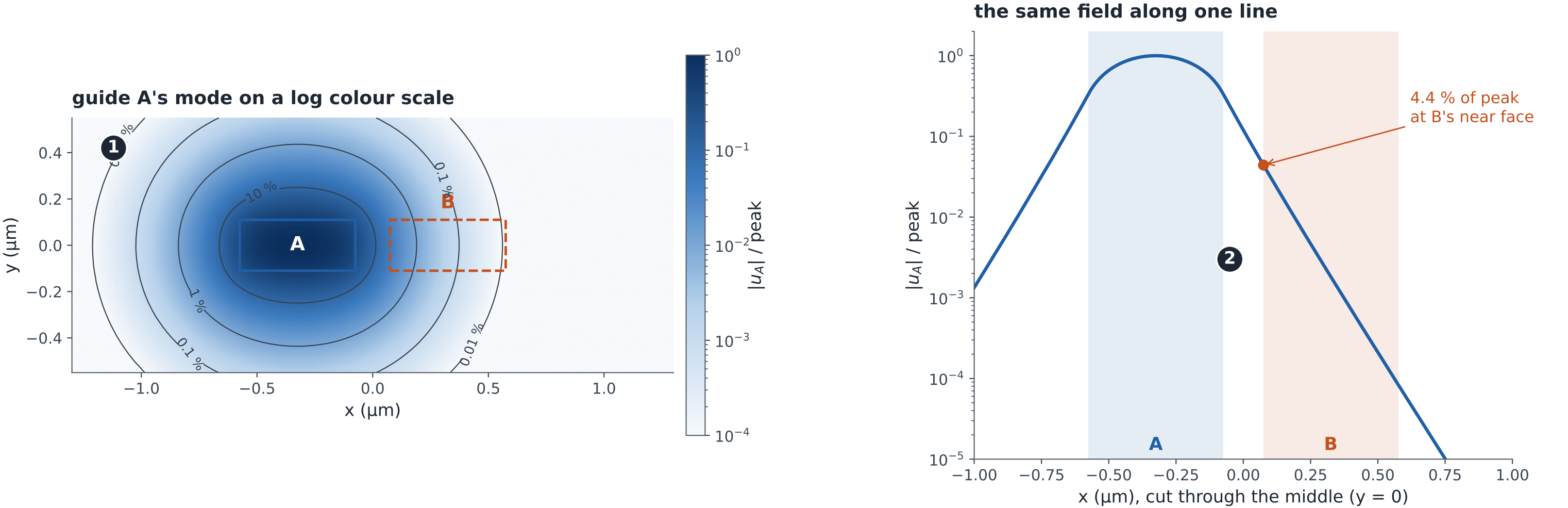


- ❶ A mask Δ is $n^2_{Si} - n^2_{ox} = 10.16$ inside its core and 0 everywhere else. u_A was solved with only the first two terms.
- ❷ Placing B adds exactly one term, Δ_B , which is non-zero only inside B's rectangle.

so → the wave equation is unchanged everywhere except inside B. Does A's field reach that far?

Does A's field actually reach B's silicon?

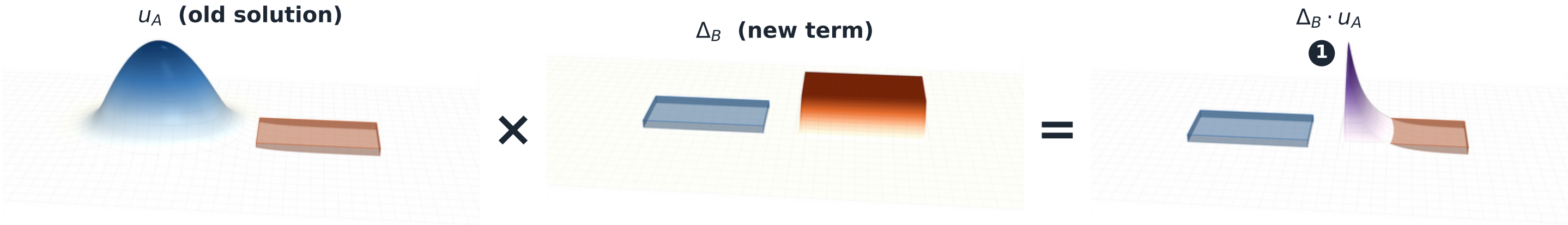
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- 1 Every contour is 10x weaker than the one inside it. The faint tail plainly overlaps B's rectangle (dashed).
- 2 A straight line on a log axis is an exponential: between the cores the field falls by e every 74 nm.
- so → part of A's field sits in B's silicon (0.023 % of $|u_A|^2$), a region A's own equation treated as silica.

Where exactly does u_A stop being a solution?

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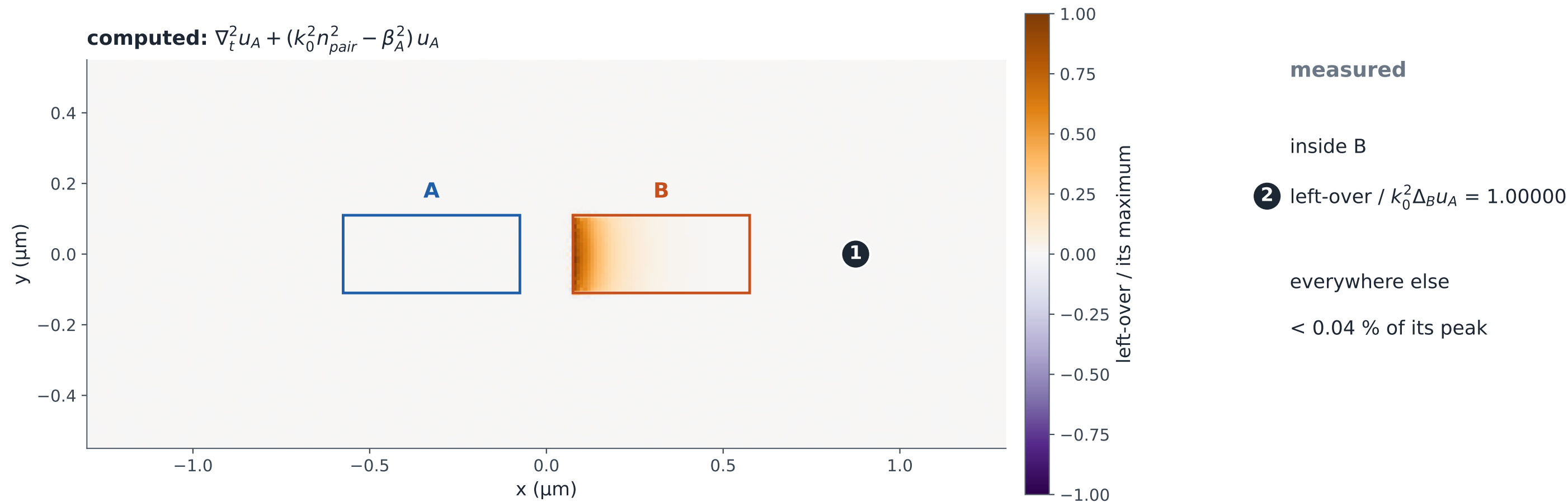


1 Multiplying by the mask keeps only the part of A's field inside B's silicon. It is tiny: drawn 24x taller than the left plot.

so → if u_A fails anywhere, it can only be here. Next page: check that on the computer.

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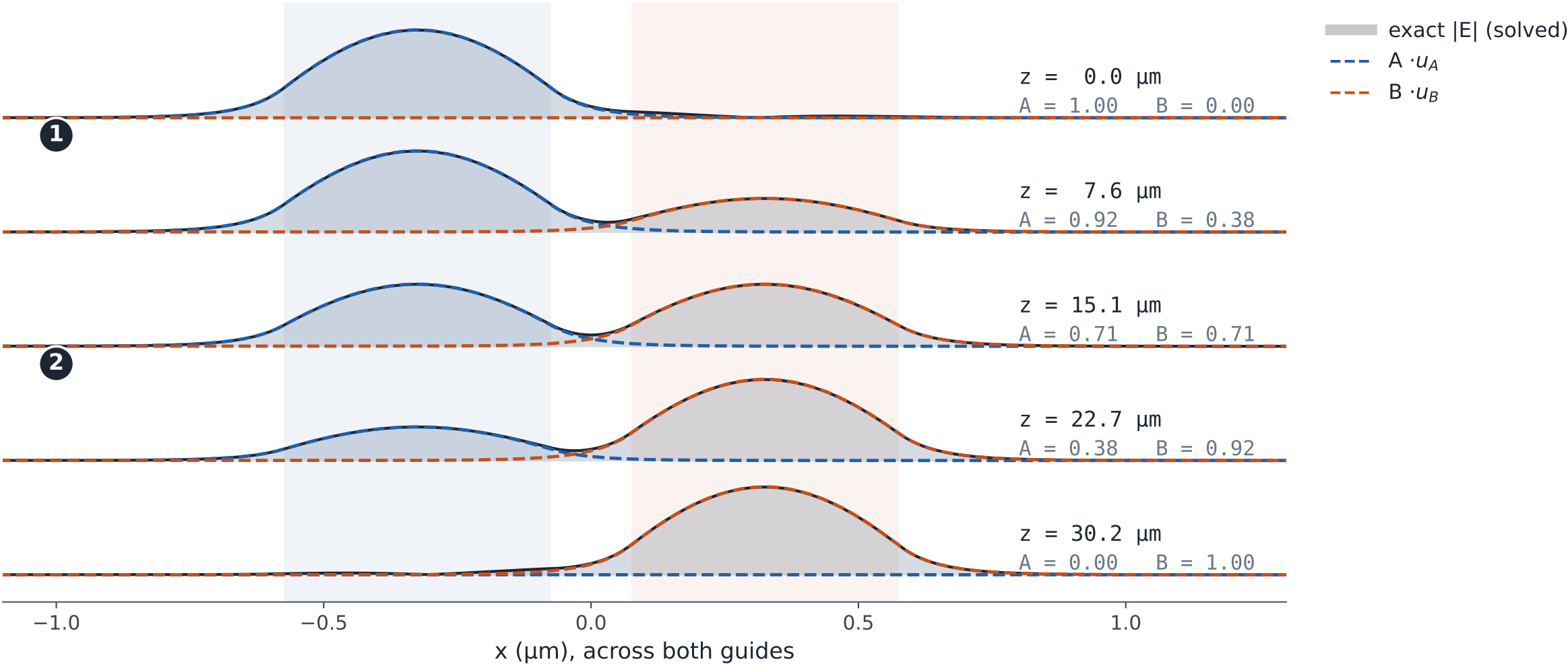
- ① Not drawn from a formula: u_A was put into the two-guide equation and what does not cancel was plotted.
- ② The left-over is exactly $k_0^2 \Delta_B u_A$: the 'source term' of the notes, located only in B.

so → a left-over means the field cannot stay as it is. It is forced to change. Into what?

If the field must change, what does it change into?

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the exact field at five positions along the coupler



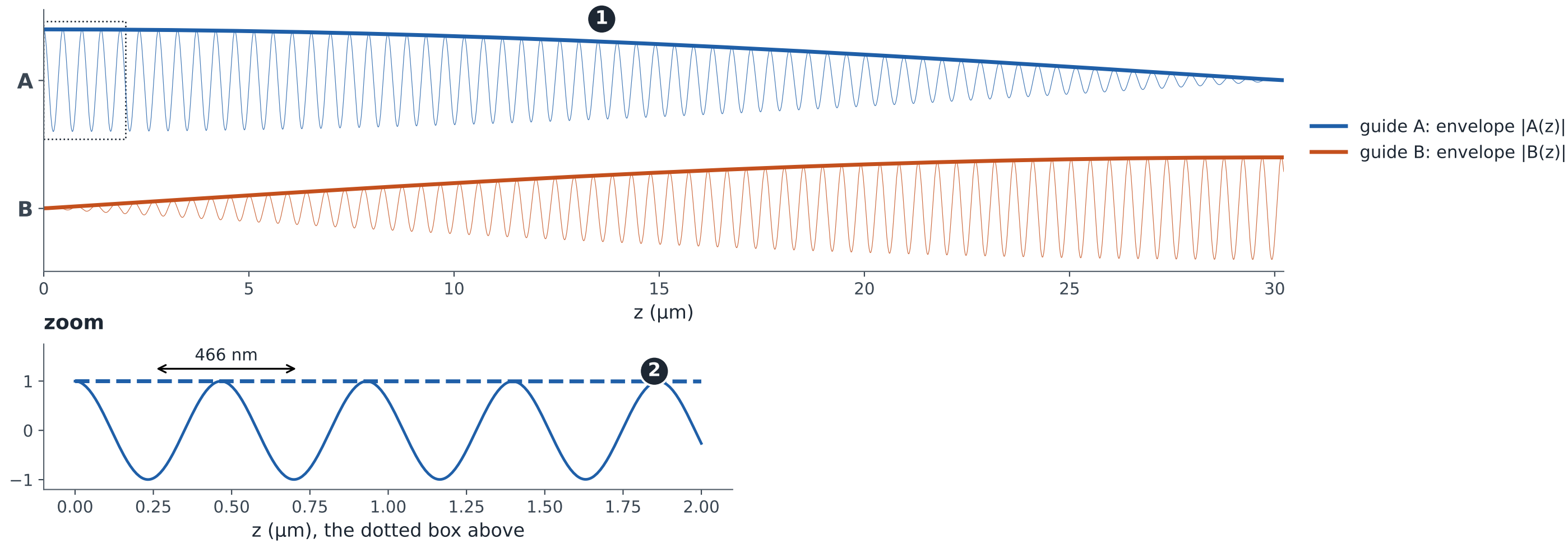
- 1 Grey: the exact field of the two-guide structure. Dashed: guide A's shape and guide B's shape, scaled by A and B.
- 2 They match at every position: the shapes stay put, only the two weights A and B change.

so → the field is $[A(z) u_A + B(z) u_B] e^{-j\beta z}$: two numbers per position describe it.

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field at each core centre along the guide: fast oscillation under a slow envelope

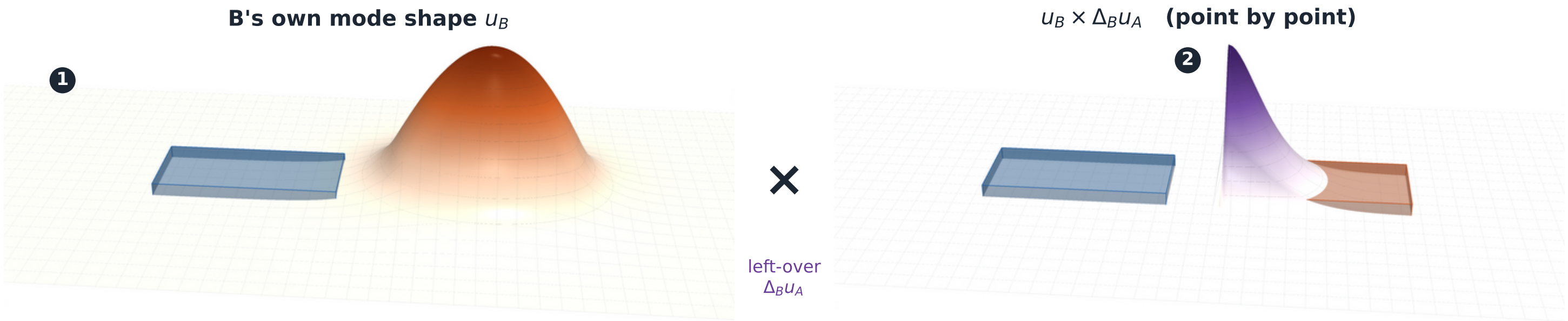


- 1 The envelopes change over 30 μm .
- 2 The carrier repeats every 466 nm, 65x faster.

so $\rightarrow$ **A and B vary slowly, so $|A''| \ll |2\beta A'|$ (ratio $\kappa/2\beta = 0.0019$): drop A'' , keep first-order equations in A and B.**

How much of A's left-over can drive B's mode?

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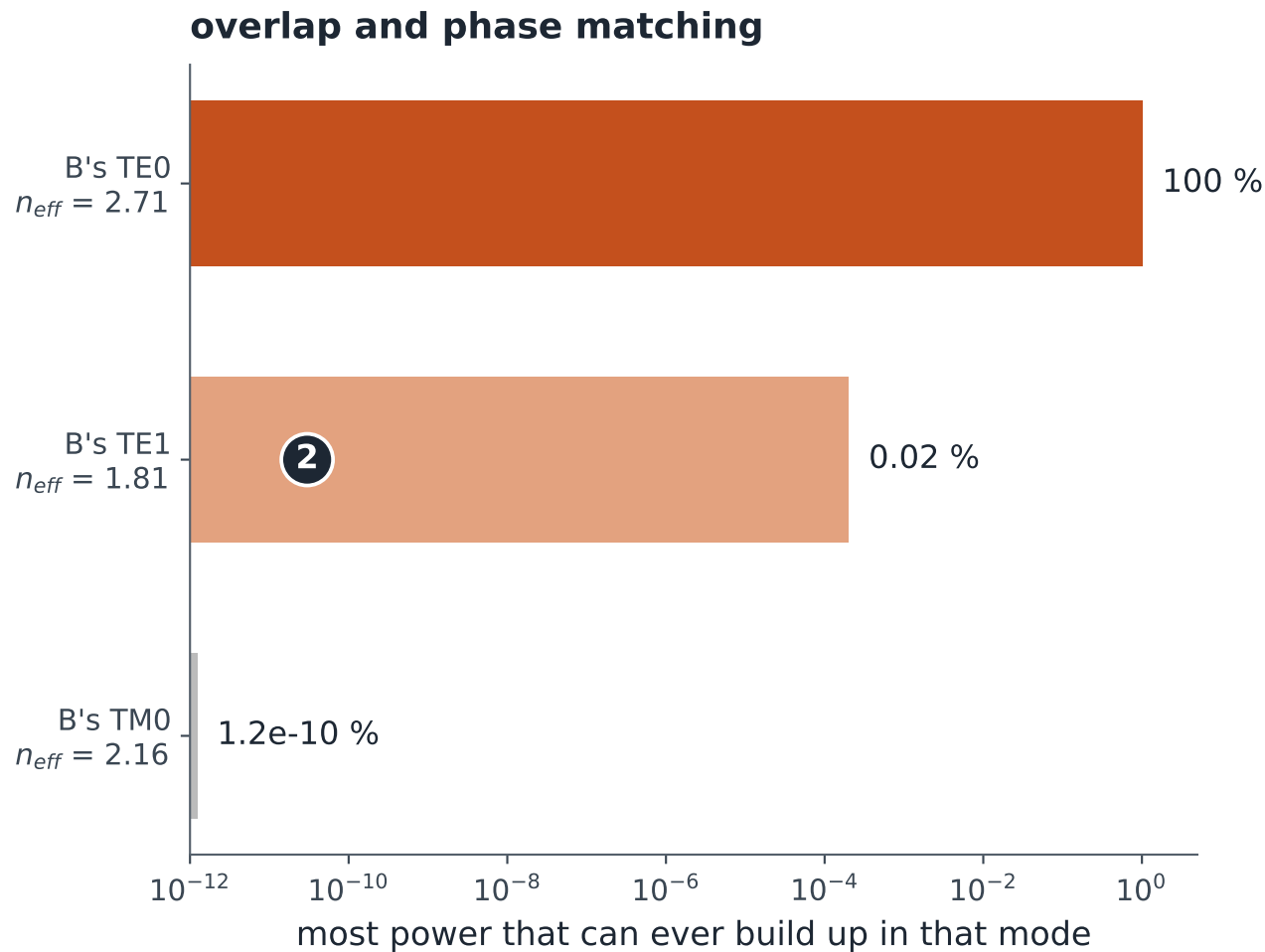
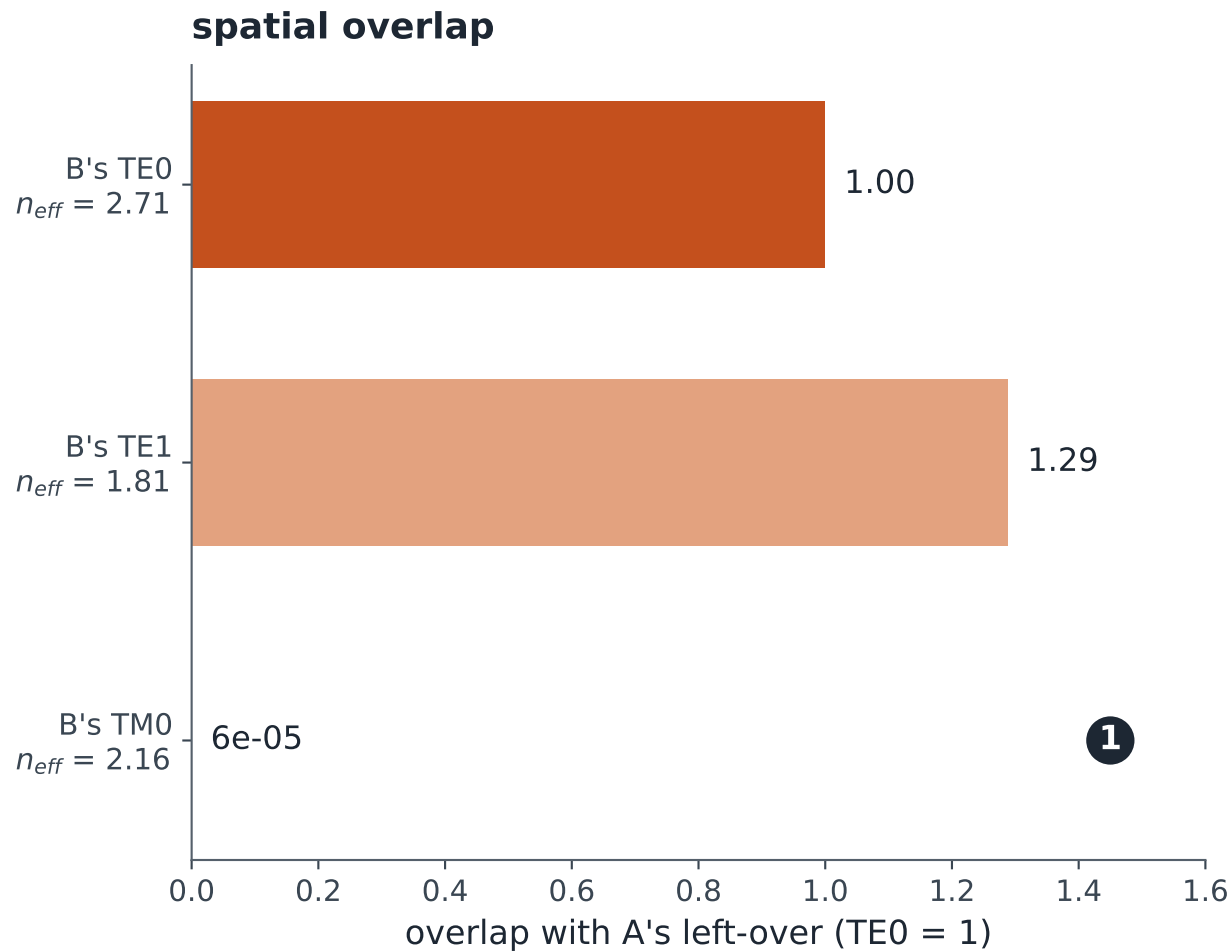


- 1 Projection asks 'how much of the left-over looks like B's mode?', like a dot product $\hat{y} \cdot \mathbf{v}$.
- 2 Multiply point by point and add up: the volume under this surface is the overlap integral, one number.

so $\rightarrow B' = -j\kappa_c A$, with $\kappa_c = \frac{k_0^2}{2\beta} \iint \Delta_B u_B u_A dA / \iint u_B^2 dA$.

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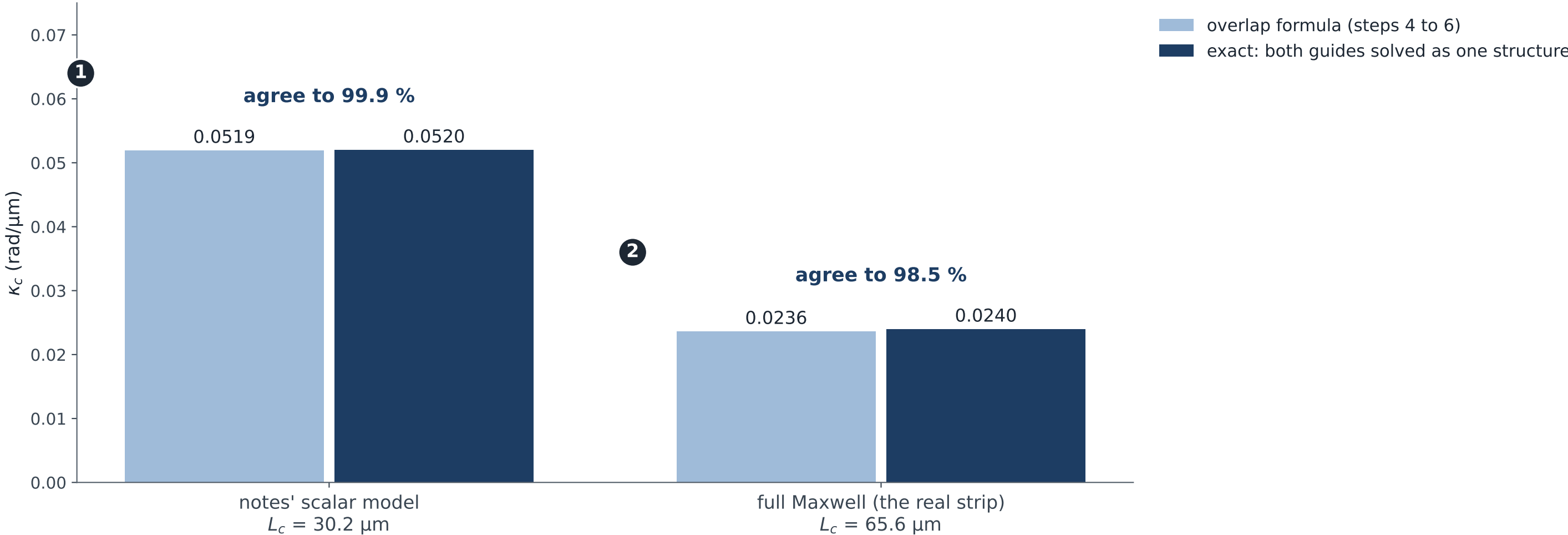


- ❶ A's lopsided tail overlaps B's TE1 even better than TE0 (1.29x). TM0 (wrong field direction) hardly at all.
- ❷ But TE1 travels at $n_{eff} = 1.81$, not 2.71: its drive slips out of step every 1.5 μm and cancels. At most 0.02 % builds up.

so → coupling needs both spatial overlap and equal β (phase matching). Identical guides have both.

What is K_C , and can the formula be trusted?

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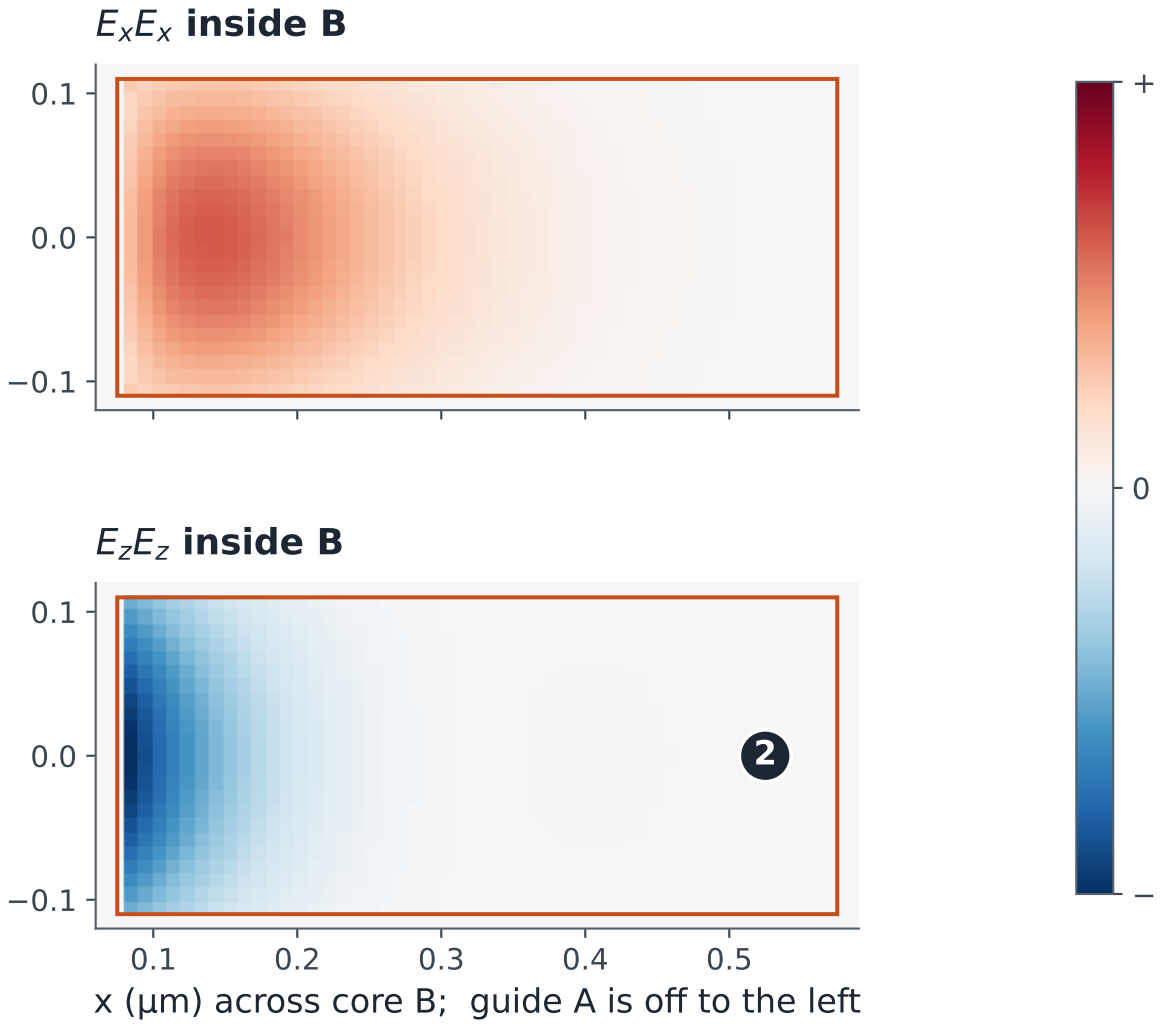
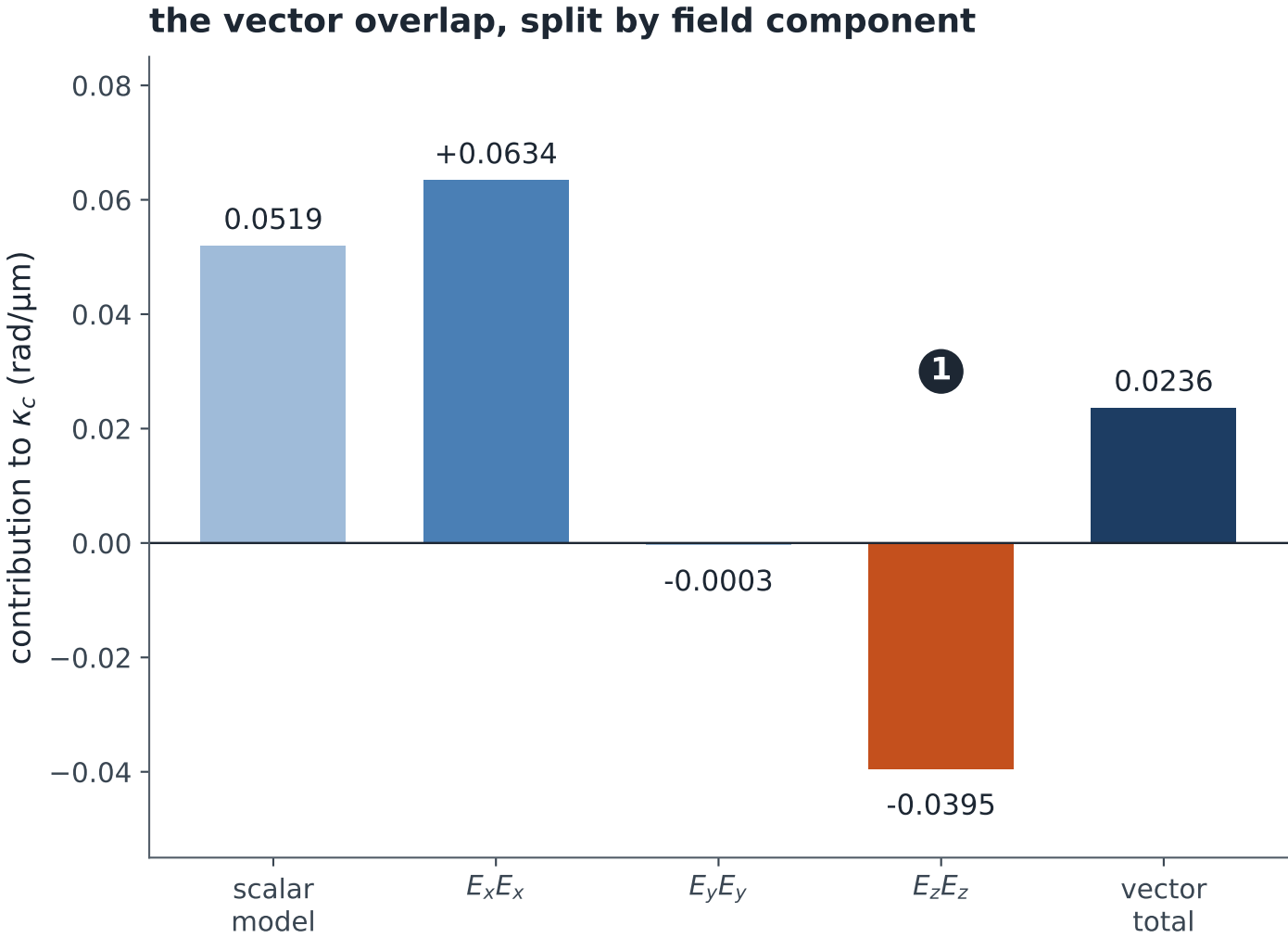


- ❶ Inside its own (scalar) model the notes' formula is right: 0.05192 against the exact 0.05198 rad/ μm .
- ❷ For the real strip the vector version of the same formula is right too, but the number is 2.2x smaller.

so $\rightarrow K_C = 0.0240 \text{ rad}/\mu\text{m}$ for this coupler. Why is the scalar model so far off?

Why is the scalar model 2x too strong for silicon?

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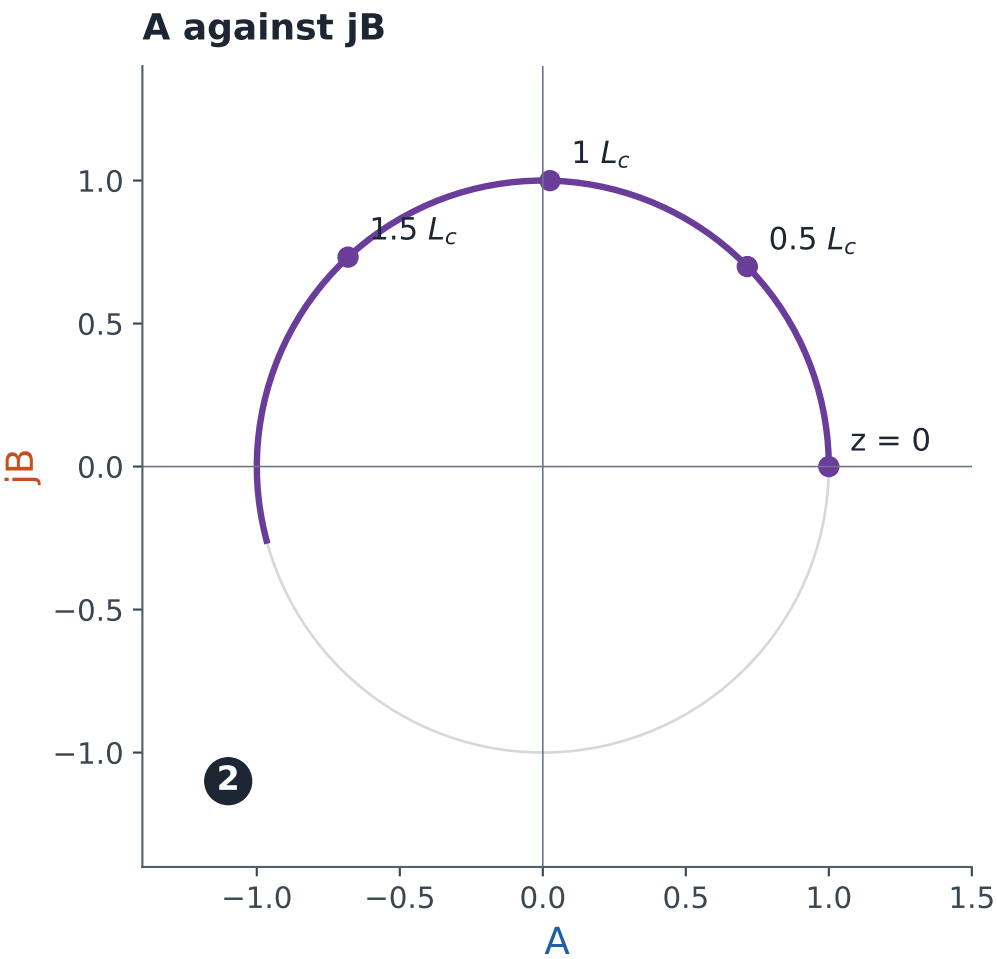
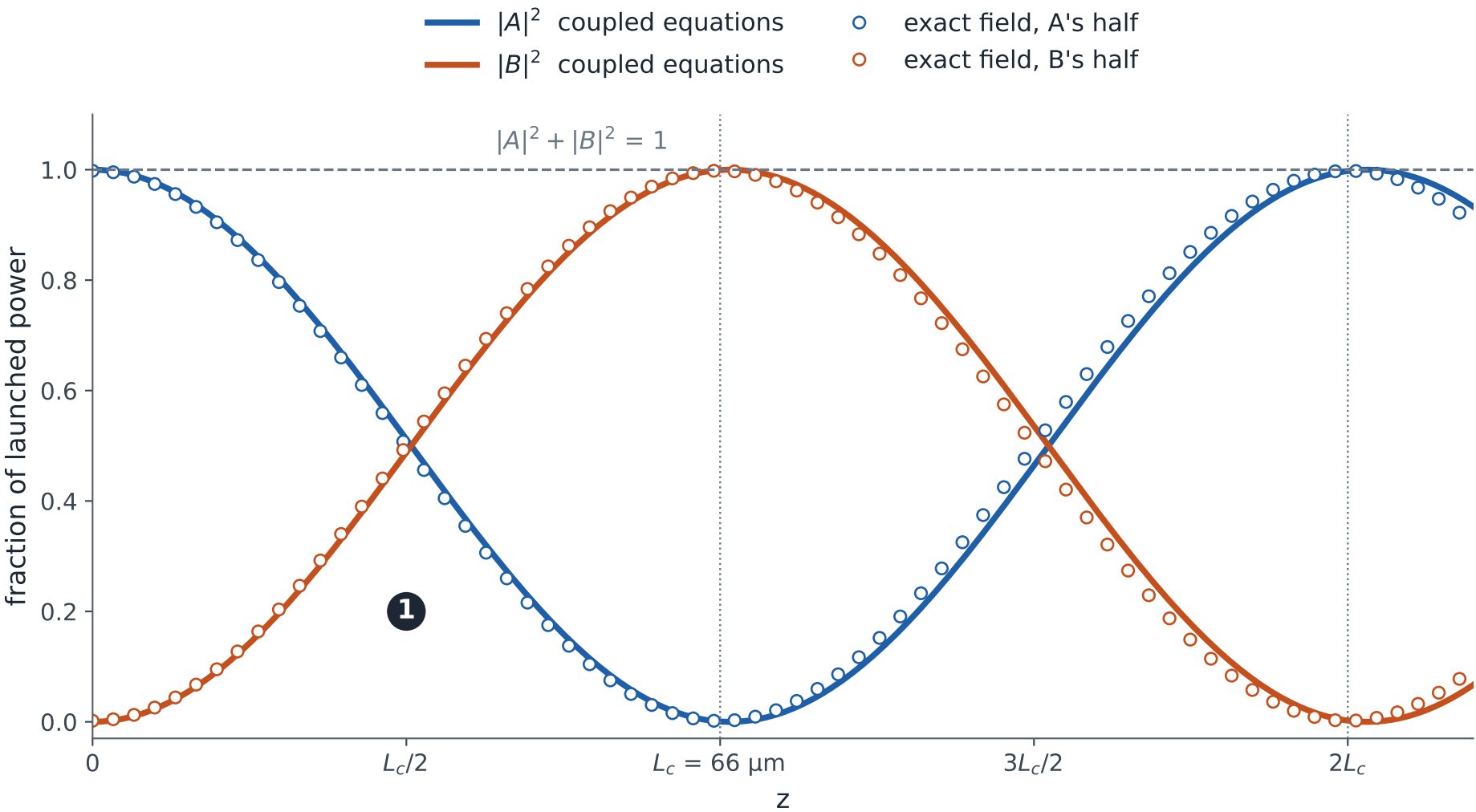


- ❶ A silicon mode's field lines tilt along z (E_z , 17 % of $|E|^2$). The scalar model has no E_z .
- ❷ Inside B, A's tilt and B's tilt point opposite ways: the $E_z E_z$ term is negative and cancels most of $E_x E_x$.

so → same logic as steps 4 to 6, but the integrand is the full vector dot product. From here on: full Maxwell.

What do $dA/dz = -j\kappa_c B$ and $dB/dz = -j\kappa_c A$ do over distance?

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- ❶ Lines: the coupled equations solved from $A = 1, B = 0$. Circles: the exact field. All power is in B after $L_c = \pi/2\kappa_c = 65.6 \mu\text{m}$.
- ❷ The state runs round a circle at rate κ_c : the oscillator $A'' = -\kappa_c^2 A$. The $-j$ means B starts 90° behind A.

so $\rightarrow P_A = \cos^2 \kappa_c z, P_B = \sin^2 \kappa_c z$, and $P_A + P_B = 1$: **power moves, it is never lost.**

Why does the power hop back and forth at all?

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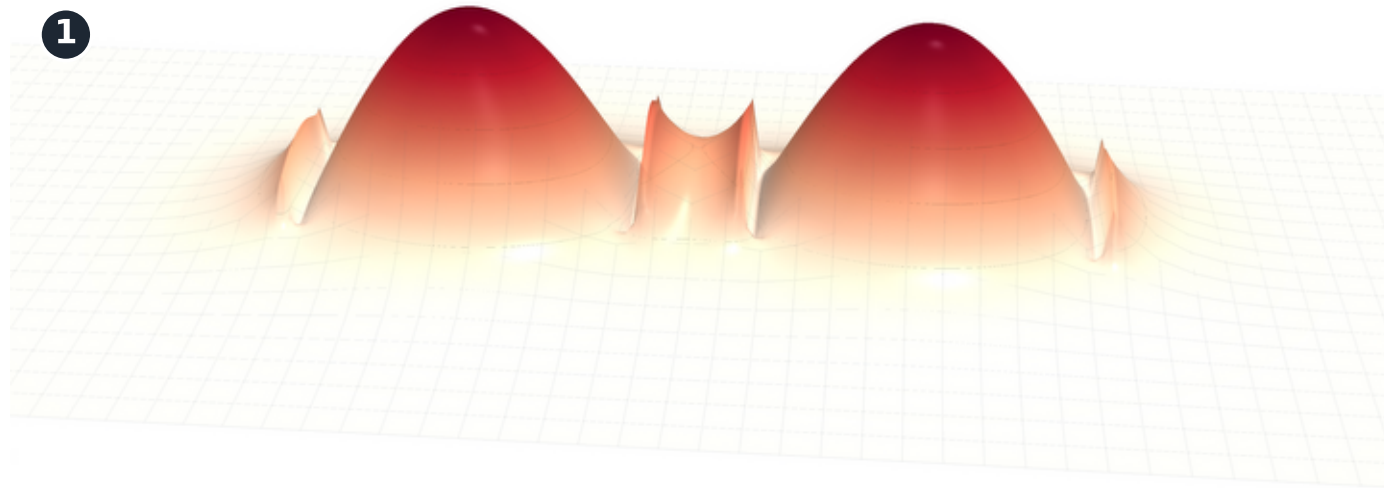
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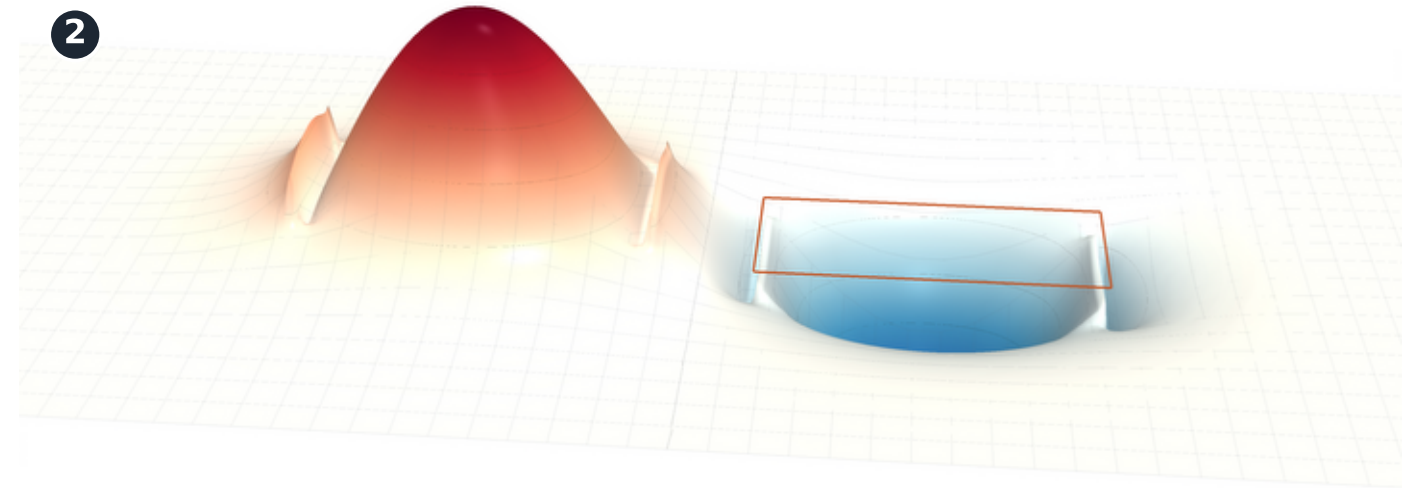
even supermode e_+ $n_{eff} = 2.7177$

1



odd supermode e_- $n_{eff} = 2.7077$

2



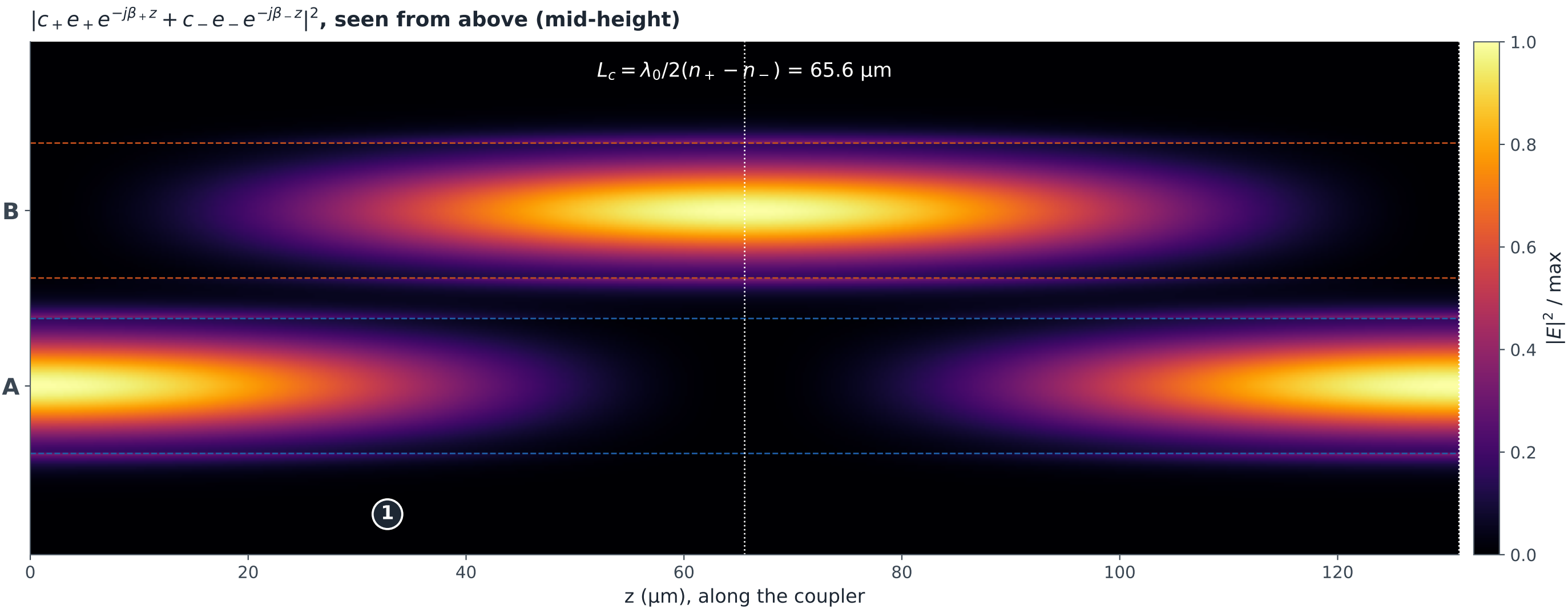
1 The true modes of the pair (field E_x ; red +, blue -). Neither lives in one guide. The spikes at the sidewalls are the normal-E jump.

2 Light launched in A alone is 50.9 % e_+ + 49.1 % e_- : they add in A and cancel in B.

so → the two supermodes travel at slightly different speeds ($n_+ - n_- = 0.0100$).

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1 Where the two supermodes are in step the light is in A; half a slip later it is in B.

so $\rightarrow$ 'power hopping' is two supermodes beating, and $(\beta_+ - \beta_-)/2$ is the same κ_c as step 7.

Does a full 3-D Maxwell simulation agree?

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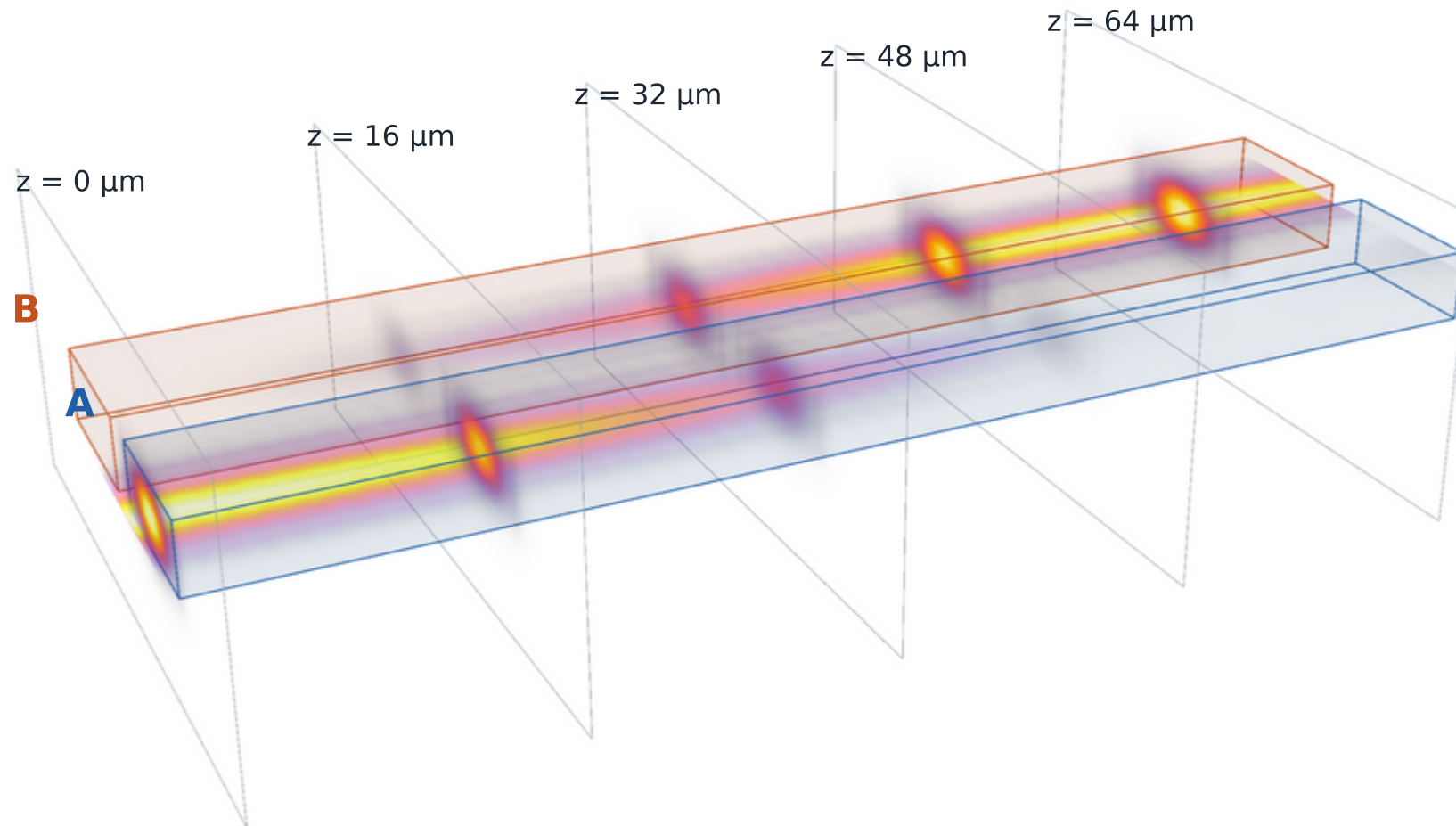
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Meep 3-D FDTD: $|E|^2$ on the mid-height plane and five cross-sections (z drawn 18x shorter)

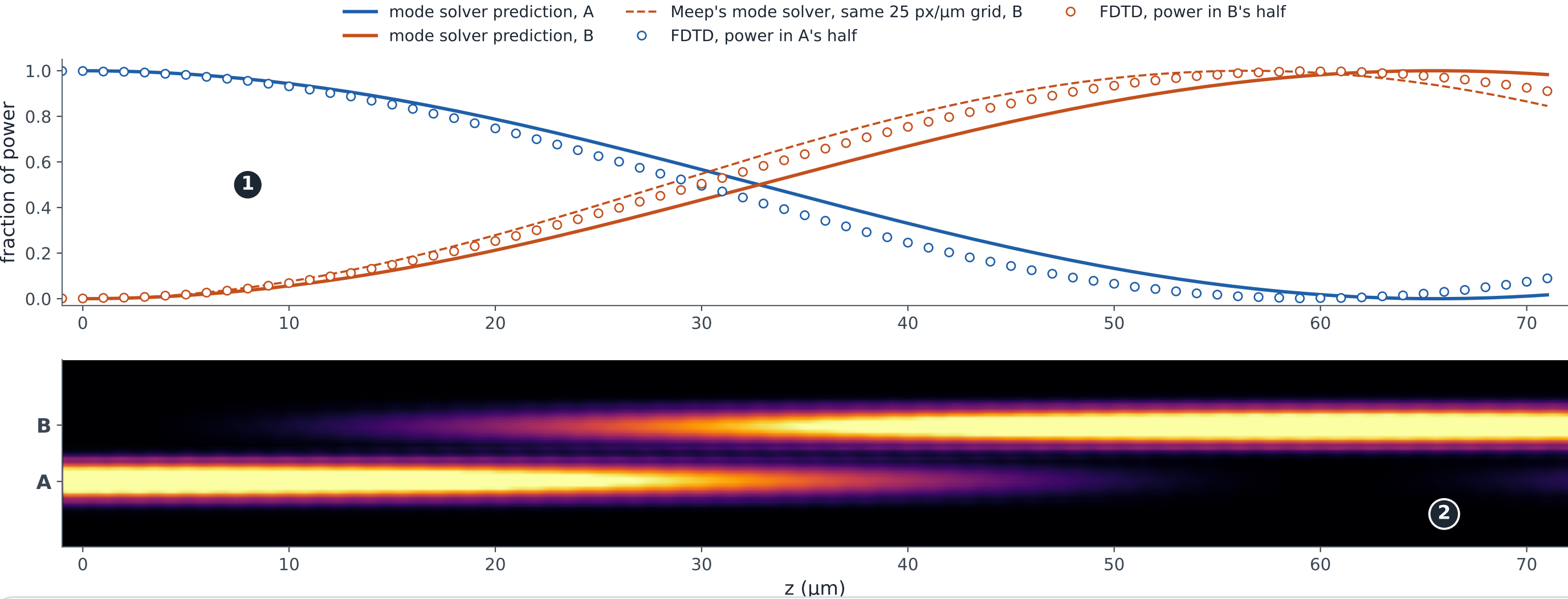


- ❶ Maxwell's equations stepped in time on a 3-D grid (25 px/ μm). No coupled-mode theory goes in.
- ❷ Light enters guide A. Guide B starts at $z = 0$, so $A = 1$, $B = 0$ there. Follow the bright core from slice to slice.

so → a brute-force Maxwell solve moves the light from A to B on its own. Next page: by how much, and how fast?

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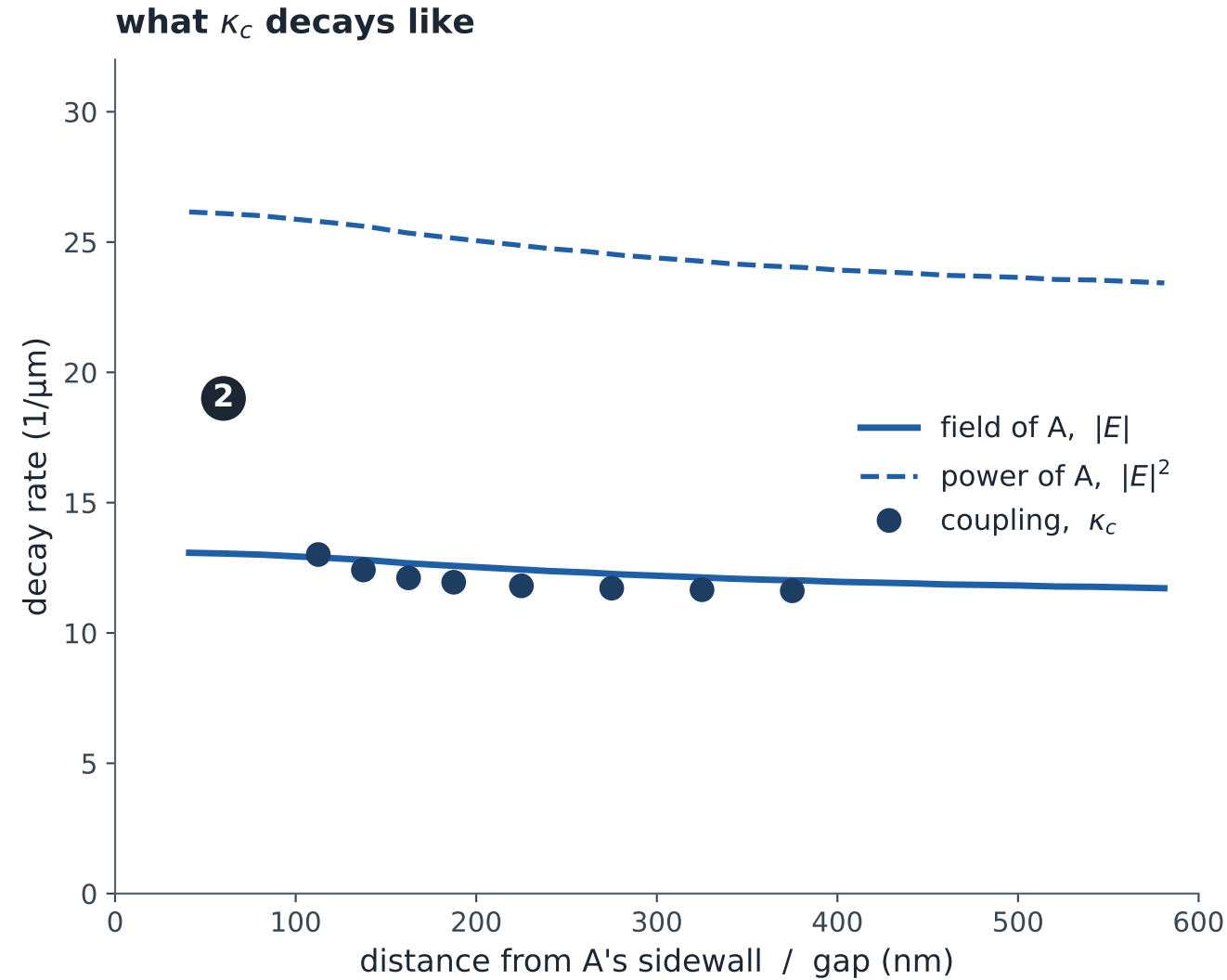
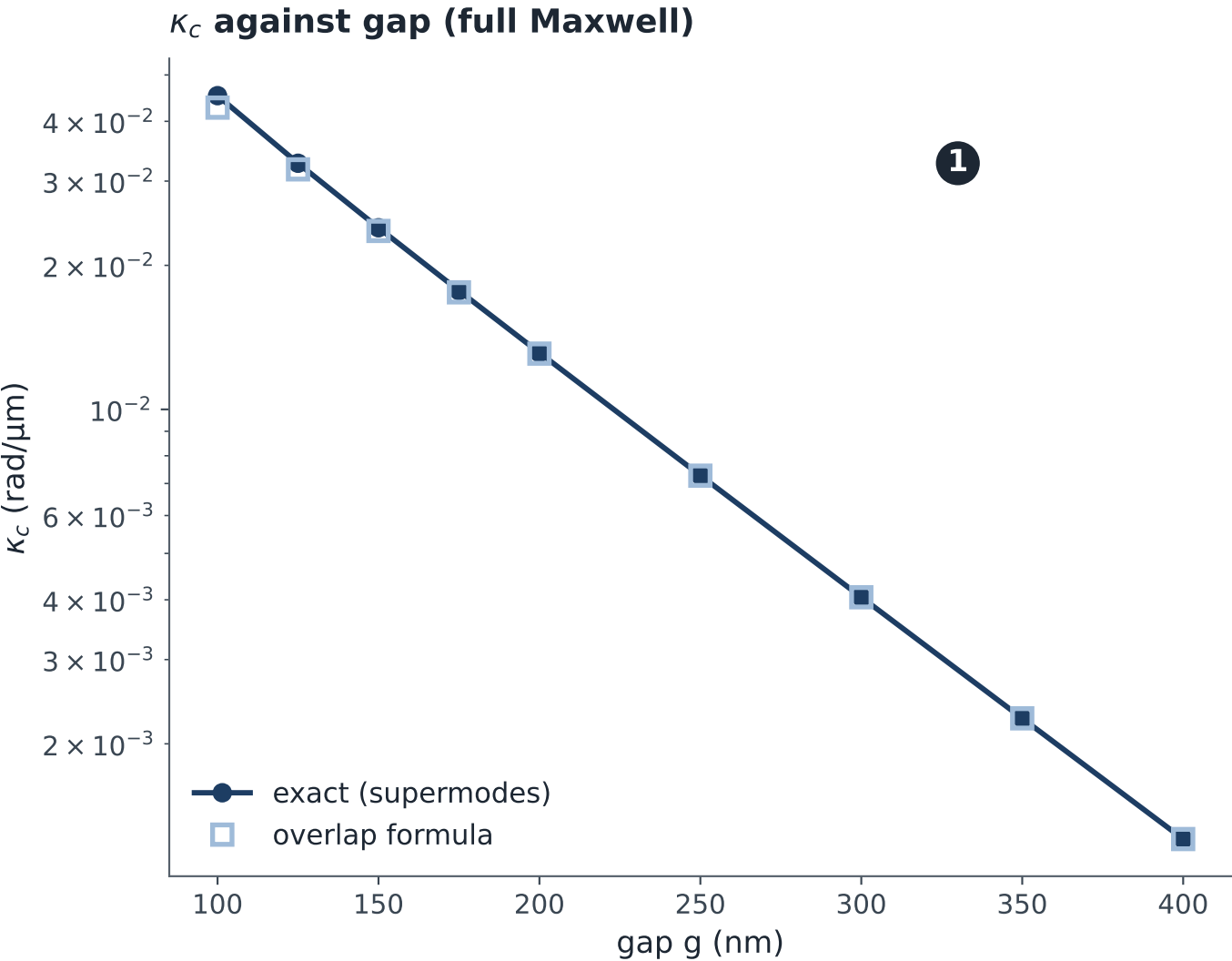


- 1 FDTD L_c at 20 px/ μm : 65.0 μm , 25 px/ μm : 59.6 μm . Converged mode solver: 65.6 μm . The FDTD grid is not converged yet.
- 2 FDTD at 25 px/ μm (dots) tracks Meep's own mode solver on the same grid (dashed) to 95 %; $P_A + P_B$ is constant to 0.01 %.

so → with no coupled-mode theory, Maxwell's equations move 99.8 % of the light into B. The exact length is grid-limited here.

Why does κ_C fall like $e^{-\gamma g}$, and not $e^{-2\gamma g}$?

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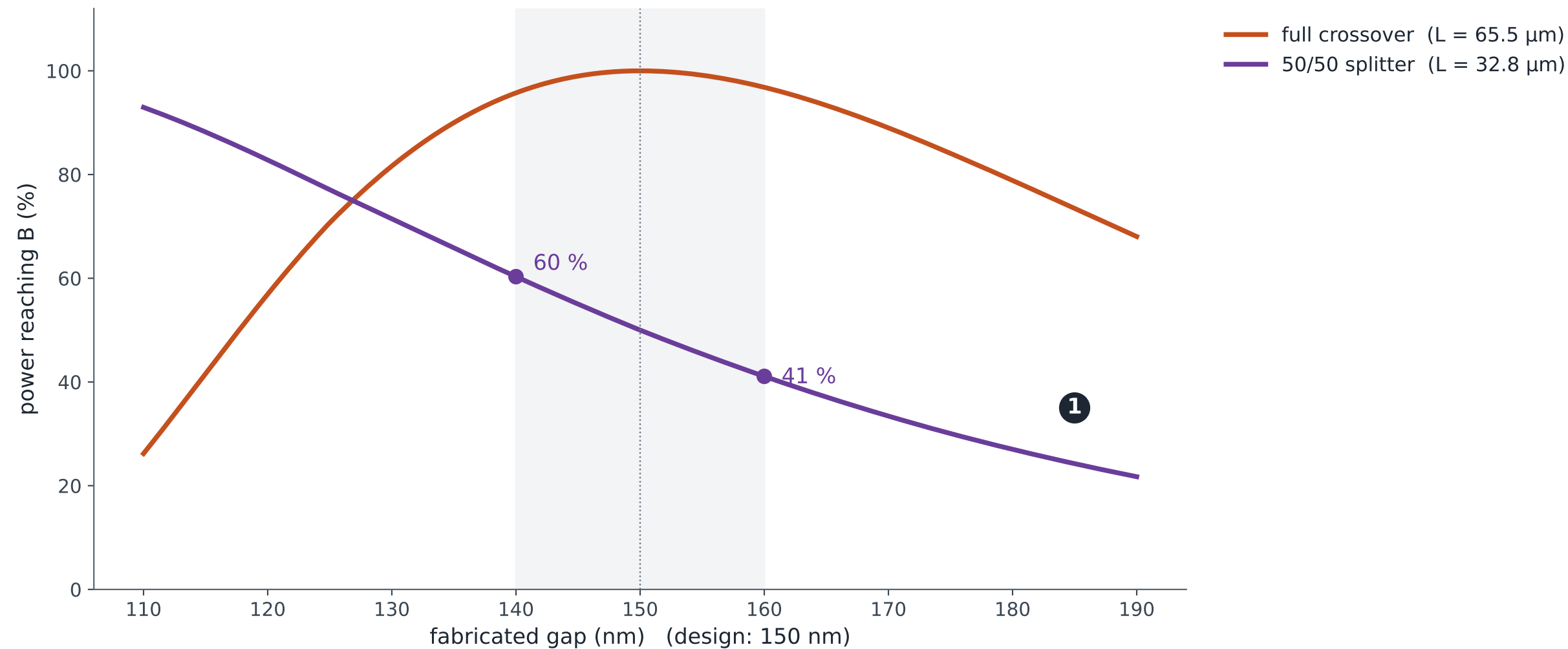


- 1 On a log axis κ_C is a straight line in gap: $\kappa_C \propto e^{-\gamma g}$, $\gamma = 11.9 / \mu\text{m}$. The overlap formula tracks it at every gap.
- 2 κ_C decays at the field's rate (solid), not the power's (dashed): the overlap integral contains u_A once.

so → one extra decay length of gap ($1/\gamma = 84 \text{ nm}$) divides κ_C by $e = 2.7$.

How much does a fabrication error in the gap matter?

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1 ± 10 nm of gap changes κ_c by +13 % / -11 %. The 50/50 splitter becomes 60/40 or 41/59; the full crossover barely moves.

so → coupler gaps are among the most tightly controlled dimensions on a chip.